DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II Advanced Calculus (2AC)

Tutorial Sheet 13: Solutions **18th and 19th May 2023**

1. We first summarize in this section, some of the main identities established or used in the lectures. Let $f:\mathbb{R}^3\to\mathbb{R}$ be a continuous function and $\vec{F}:\mathbb{R}^3\to\mathbb{R}^3$ be a (continuous) vector field. Let $\mathcal{C}$ be a regular curve described by a vecor function $\vec{r}:[a,b]\to\mathbb{R}^3$. Then we recall that

$$\left\{\begin{array}{l}
\underbrace{\int_{\mathcal{C}} f(x,y,z)ds=\int_a^b f(\vec{r}(t))\,\|\vec{r}\,'(t)\|\;dt}_{\text{Line integral of scalar-valued function}}\,, \qquad \underbrace{\int_{\mathcal{C}} \vec{F}\cdot d\vec{r}=\int_a^b \vec{F}(\vec{r}(t))\cdot\vec{r}\,'(t)\,dt}_{\text{Line integrals of a vector field}} \\[2ex]
\underbrace{\int_{\mathcal{C}} \nabla f\cdot d\vec{r}=\int_a^b \nabla f(\vec{r}(t))\cdot\vec{r}\,'(t)\,dt=\int_a^b \frac{d(f\circ\vec{r})}{dt}\,dt=f(\vec{r}(b))-f(\vec{r}(a))}_{(f\text{ of class }C^1!\text{ Fundamental Theorem of Calculus for line integrals})}\,, \\[2ex]
\underbrace{\int_{\mathcal{C}}=\sum_{i=1}^n\int_{\mathcal{C}_i}}_{c\text{ piecewise regular}} \quad \text{when } \mathcal{C}=\bigcup_{i=1}^n\mathcal{C}_i \text{ with } \mathcal{C}_i\cap\mathcal{C}_j=\{\text{single point at most}\} \text{ for } i\neq j \\[2ex]
(*)\ \int_{-\mathcal{C}} fds=\int_{\mathcal{C}} fds \quad \text{while} \quad (**)\ \int_{-\mathcal{C}}\vec{F}\cdot d\vec{r}=-\int_{\mathcal{C}}\vec{F}\cdot d\vec{r} \\
-\mathcal{C} \text{ denotes the curve having the same points as } \mathcal{C} \text{ with the opposite orientation}
\end{array}\right.$$

We want to explain why $(*)$ and $(**)$ hold for a regular curve $\mathcal{C}$ in $\mathbb{R}^2$ and $f:\mathbb{R}^2\to\mathbb{R}$).

(1) Let $\vec{r}:[a,b]\to\mathbb{R}^2$ be a vector function describing $\mathcal{C}$ and deduce a vector function $\vec{u}:[a,b]\to\mathbb{R}^2$ describing the curve $-\mathcal{C}$.

$$\vec{u}(t):=\vec{r}(a+b-t)\quad t\in[a,b]\,.$$

So,

$$\vec{u}\,'(t)=-\vec{r}\,'(a+b-t)\,.$$

Therefore,

$$\int_{\mathcal{C}} fds=\int_a^b f(\vec{r}(t))\,\|\vec{r}\,'(t)\|\,dt$$

and

$$\begin{aligned}\int_{-\mathcal{C}} fds &= \int_a^b f(\vec{u}(t))\,\|\vec{u}\,'(t)\|\,dt=\int_a^b f(\vec{r}(a+b-t))\,\|\vec{r}\,'(a+b-t)\|\,dt \\ &= \underbrace{\int_b^a f(\vec{r}(\tau)\,\|\vec{r}\,'(\tau)\|\,(-d\tau)}_{\text{Change of variables: }\tau=a+b-t}=\int_a^b f(\vec{r}(\tau)\,\|\vec{r}\,'(\tau)\|\,d\tau=\int_{\mathcal{C}} fds\,.\end{aligned}$$

(2) Use the vector functions $\vec{r}$ and $\vec{u}$ to evaluate $\int_{\mathcal{C}} f(x,y)ds$ and $\int_{-\mathcal{C}} f(x,y)ds$ respectively and check that $(*)$ holds.

(3) Use the vector functions $\vec{r}$ and $\vec{u}$ to evaluate $\int_{\mathcal{C}} \vec{F}\cdot d\vec{r}$ and $\int_{-\mathcal{C}} \vec{F}\cdot d\vec{r}$ respectively and check that $(**)$ holds.
Using the vector functions $\vec{r}$ and $\vec{u}$ we get

$$\int_{\mathcal{C}} \vec{F}\cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t))\cdot \vec{r}\,'(t)dt$$

while

$$\begin{aligned}\int_{-\mathcal{C}} \vec{F}\cdot d\vec{r} &= \int_a^b \vec{F}(\vec{u}(t))\cdot \vec{u}\,'(t)dt = \int_a^b \vec{F}(\vec{r}(a+b-t))\cdot(-\vec{r}\,'(a+b-t))dt \\ &= \underbrace{\int_b^a \vec{F}(\vec{r}(\tau))\cdot \vec{r}\,'(\tau)d\tau}_{\text{Change of variables: } \tau=a+b-t} = -\int_a^b \vec{F}(\vec{r}(\tau))\cdot \vec{r}\,'(\tau)d\tau = -\int_{\mathcal{C}} \vec{F}\cdot d\vec{r}.\end{aligned}$$

2. Let $\vec{F}:\mathbb{R}^2\to\mathbb{R}^2$ be the vectpor field defined by $\vec{F}(x,y)=(2x^2y^3, 3x^2y^2)$ for every $(x,y)\in\mathbb{R}^2$.

(a) Use the Green theorem to evaluate the line integral $\oint_C \vec{F}\cdot d\vec{r}$ where C is the piecewise regular curve in the first quadrant with an anticlockwise orientation, formed by the x-axis, the line $x=1$ and and the curve $y=x^2$.
Let D be the region enclosed by the curve C. By the Green Theorem, we find that

$$\begin{aligned}\int_C \boldsymbol{F}\cdot d\boldsymbol{r} &= \iint_D \left(\frac{\partial(3x^2y^2)}{\partial x} - \frac{\partial(2x^2y^3)}{\partial y}\right) = \int_0^1\int_0^{x^2}(6xy^2-6x^2y^2)dydx \\ &= \int_0^1 \Big[2xy^3-2x^2y^3\Big]_0^{x^2}dx = \int_0^1(2x^7-2x^8)dx = \Big[\frac{1}{4}x^8-\frac{2}{9}x^9\Big]_0^1 = \frac{1}{36}.\end{aligned}$$

(b) Evaluate $\oint_C \vec{F}\cdot d\vec{r}$ directly. (Hint: Use the formula for the line integral on a piecewise regular curve. So, write $C=C_1\cup C_2\cup C_3$, and find a suitable vector function for each arc of curve C_1, C_2, C_3 taking into account the fact C has a counterclockwise orientation.)

The curve C with an anticlockwise orientation is made of the arcs:

(i) C_1 which is the segment line on the x-axis from $(0,0)$ to $(1,0)$, and is described by the vector function $\vec{r}_1(t)=(t,0)$ with $0\le t\le 1$;

(ii) C_2 which is the segment line from $(1,0)$ to $(1,1)$ described by the vector function $\vec{r}_2(t)=(1,t)$ with $0\le t\le 1$.

(ii) C_3 which is the arc of the parabola $y=x^2$ from $(1,1)$ to $(0,0)$, and is described by the vector function $\vec{r}_3(t)=(1-t,(1-t)^2)$ with $0\le t\le 1$.

So,

$$\boxed{\oint_C \vec{F}\cdot d\vec{r} = \int_{C_1}\vec{F}\cdot d\vec{r} + \int_{C_2}\vec{F}\cdot d\vec{r} + \int_{C_3}\vec{F}\cdot d\vec{r}}. \qquad (*)$$

We find that

$$\int_{C_1}\vec{F}\cdot d\vec{r} = \int_0^1 (0,0)\cdot(1,0)dt = 0,$$

$$\int_{C_2} \vec{F} \cdot d\vec{r} = \int_0^1 (2t^3, 3t^2) \cdot (0,1) dt = \int_0^1 3t^2 dt = \left[t^3\right]_0^1 = 1,$$

$$\begin{aligned}
\int_{C_3} \vec{F} \cdot d\vec{r} &= \int_0^1 (2(1-t)^8, 3(1-t)^6) \cdot (-1, -2(1-t)) dt \\
&= \int_0^1 \left[-2(1-t)^8 - 6(1-t)^7\right] dt = \left[\frac{2}{9}(1-t)^9 + \frac{6}{8}(1-t)^8\right]_0^1 \\
&= -\frac{2}{9} - \frac{6}{8} = -\frac{35}{36}\,.
\end{aligned}$$

Thus, we get from $(*)$ that

$$\oint_C \vec{F} \cdot d\vec{r} = 0 + 1 - \frac{35}{36} = \frac{1}{36}\,.$$

3. If $\vec{F} = (P, Q) : \mathbb{R}^2 \to \mathbb{R}^2$ is a $\mathcal{C}^1$ vector field such that $P_y(x, y) = Q_x(x, y)$ in a subset U of $\mathbb{R}^2$, then the line integral

$$\oint_{\mathcal{C}} P(x, y)dx + Q(x, y)dy = 0$$

for any piecewise regular simple closed curve $\mathcal{C}$ contained in U.
Hint: Consider both cases: (i) $\mathcal{C}$ with a counterclockwise orientation and (ii) $\mathcal{C}$ with a clockwise orientation.

(i) For $\mathcal{C}$ with a counterclockwise orientation, we apply directly the Green theorem and get

$$\oint_{\mathcal{C}} P(x, y)dx + Q(x, y)dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dxdy = \iint_D 0dxdy = 0\,.$$

(ii) For $\mathcal{C}$ with a clockwise orientation, we apply the Green theorem to the curve $-C$ and get

$$\oint_{-\mathcal{C}} P(x, y)dx + Q(x, y)dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dxdy = \iint_D 0dxdy = 0\,.$$

Now, using $(**)$ in the first box in Section 1, we get

$$\oint_{\mathcal{C}} P(x, y)dx + Q(x, y)dy = -\oint_{-\mathcal{C}} P(x, y)dx + Q(x, y)dy = 0\,.$$

4. We first summarize in this section, some of the main identities established or used in the lectures. Let $f : \mathbb{R}^3 \to \mathbb{R}$ be a continuous function and $\vec{F} : \mathbb{R}^3 \to \mathbb{R}^3$ be a (continuous) vector field. Let $\mathcal{S}$

be a regular surface described by the vector function $\vec{r}: D \subset \mathbb{R}^2 \to \mathbb{R}^3$. Then,

$$\underbrace{A(\mathcal{S}) = \iint_D \|\vec{r}_s(s,t) \times \vec{r}_t(s,t)\| \, dsdt}_{\text{The area of the surface } \mathcal{S}}$$

$$\underbrace{\iint_{\mathcal{S}} f(x,y,z)dS = \iint_D f(\vec{r}(s,t)) \, \|\vec{r}_s(s,t) \times \vec{r}_t(s,t)\| \, dsdt}_{\text{The surface integral of the function } f \text{ over the surface } \mathcal{S}}$$

$$\underbrace{\iint_{\mathcal{S}} \vec{F}(x,y,z) \cdot \vec{n} dS = \iint_D \vec{F}(\vec{r}(s,t)) \cdot [\vec{r}_s(s,t) \times \vec{r}_t(s,t)] \, dsdt}_{\text{The flux integral of the vector field } \vec{F} \text{ across the surface } \mathcal{S}}$$

$$\underbrace{\iint_{\mathcal{S}} f(x,y,z)dS = \sum_{i=1}^{n} \iint_{\mathcal{S}_i} f(x,y,z)dS \quad \text{and} \quad \iint_{\mathcal{S}} \vec{F} \cdot \vec{n} dS = \sum_{i=1}^{n} \iint_{\mathcal{S}_i} \vec{F} \cdot \vec{n} dS}_{\text{In case } \mathcal{S} \text{ is piecewise regular: } \mathcal{S} = \bigcup_{i=1}^{n} \mathcal{S}_i \text{ with each } \mathcal{S}_i \text{ regular and } \mathcal{S}_\rangle \cap \mathcal{S}_j \subset \{\text{a curve}\} \text{ for } i \neq j}$$

(a) Show that $\displaystyle\iint_S xydS = 4\int_0^1 \int_0^{\frac{\pi}{2}} r^3 \cos\theta \sin\theta \sqrt{r^2(4+5\cos^2\theta)+1} d\theta dr$ where the surface S described by the vector function $\vec{r}(s,t) = (s, 2t, s^2+3t^2)$ with $s^2+t^2 \leq 1$, $s \geq 0$ and $t \geq 0$.

$$\iint_S xydS = \iint_D 2st|\vec{r}_s(s,t) \times \vec{r}_t(s,t)|dsdt \quad \text{where } D = \{(s,t) \colon \, s^2+t^2 \leq 1, \, s \geq 0, \, t \geq 0\}$$

$$\left.\begin{array}{l} \vec{r}_s(s,t) = (1,0,2s) \\ \vec{r}_t(s,t) = (0,2,6t) \end{array}\right\} \Rightarrow \vec{r}_s(s,t) \times \vec{r}_t(s,t) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 2s \\ 0 & 2 & 6t \end{vmatrix} = (-4s, -6t, 2)$$

So,

$$\iint_S xydS = \iint_D 2st\sqrt{16s^2+36t^2+4}dsdt = 4\int_0^1 \int_0^{\frac{\pi}{2}} r^3 \cos\theta \sin\theta \sqrt{r^2(4+5\cos^2\theta)+1} d\theta dr$$

(b) Evaluate the flux integral $\displaystyle\iint_S \vec{F} \cdot \vec{n} dS$ where $\vec{F}(x,y,z) = (z, x, y^2)$ and S is the part of the paraboloid $z = x^2+y^2$ lying inside the cylinder $x^2+y^2 = 4$ and whose boundary is oriented counterclockwise.
The unit normal field is $\vec{n} = \dfrac{(-2x,-2y,1)}{\sqrt{4x^2+4y^2+1}}$. Thus, if $D = \{(x,y) \colon \, x^2+y^2 \leq 4\}$,

$$\begin{aligned}
\iint_S \vec{F} \cdot \vec{n} dS &= \iint_D (x^2+y^2, x, y^2) \cdot \frac{(-2x,-2y,1)}{\sqrt{4x^2+4y^2+1}} \sqrt{1+4x^2+4y^2} dxdy \\
&= \iint_D [-2x(x^2+y^2) - 2xy + y^2] dxdy \\
&= \int_0^{2\pi} \int_0^2 r[-2r^3\cos\theta - 2r^2\cos\theta\sin\theta + r^2\sin^2\theta] drd\theta \\
&= 0 + 0 + \int_0^{2\pi} \int_0^2 r^3 \sin^2\theta drd\theta = \left[\int_0^2 r^3 dr\right] \left[\int_0^{2\pi} \frac{1-\cos(2\theta)}{2} d\theta\right] = 4\pi
\end{aligned}$$

5. If $\vec{F}(x,y,z) := (2x^2 - yz, xyz^2, -x - 2zy^2)$ then find $\text{div}(\vec{F})$ and curl$F$.
We recall that if $\vec{F} = (P,Q,R)$, then

$$\text{div}(\vec{F}) = (\partial/\partial x, \partial/\partial y, \partial/\partial z) \cdot (P,Q,R) = P_x + Q_y + R_z = 4x + xz^2 - 2y^2 .$$

and

$$\begin{aligned} \text{curl}(\vec{F}) &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} = (R_y - Q_z, P_z - R_x, Q_x - P_y) \\ &= (-4yz - 2xyz, -y + 1, yz^2 + z) . \end{aligned}$$

6. (a) Verify the Stokes theorem for the vector field $F(x,y,z) = (y^2, z^2, y^2)$ and S being the part of the plane $x + 2y + 3z = 6$ lying in the first octant.
By Stokes theorem

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl}(\vec{F}) \cdot \vec{n} dS$$

with C being the union of the segment lines

$C_1 :\ x+2y = 6$ in the xy-plane, $C_2 :\ x+3z = 6$ in the xz-plane, $C_3 :\ 2y+3z = 6$ in the yz-plane.

$$\oint_{C_1} \vec{F} \cdot d\vec{r} = \int_0^3 (y^2, 0, y^2) \cdot (-2, 1, 0) dy = \int_0^3 -2y^2 dy = -18 .$$

$$\oint_{C_2} \vec{F} \cdot d\vec{r} = \int_2^0 (0, z^2, 0) \cdot (-3, 0, 1) dz = 0 .$$

$$\oint_{C_3} \vec{F} \cdot d\vec{r} = \int_3^0 (y^2, (2 - \frac{2}{3}y)^2, y^2) \cdot (0, 1, -\frac{2}{3}) dy = -\int_0^3 (4 - \frac{8}{3}y - \frac{2}{9}y^2) dy = 2 .$$

Thus $\oint_C \vec{F} \cdot d\vec{r} = \oint_{C_1} \vec{F} \cdot d\vec{r} + \oint_{C_2} \vec{F} \cdot d\vec{r} + \oint_{C_3} \vec{F} \cdot d\vec{r} = -18 + 0 + 2 = -16 .$

Now to evaluate $\iint_S \text{curl}(\vec{F}) \cdot \vec{n} dS$ we need

$$\text{curl}(\vec{F}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 & z^2 & y^2 \end{vmatrix} = (2y - 2z, 0, -2y) \quad \text{and} \quad \vec{n} = \frac{(1,2,3)}{\sqrt{14}}$$

Hence,

$$\begin{aligned} \iint_S \text{curl}(\vec{F}) \cdot \vec{n} dS &= \iint_D \frac{1}{\sqrt{14}} (-4y - 4 + \frac{2}{3}x + \frac{4}{3}y) \frac{\sqrt{14}}{3} dxdy \\ &= \frac{1}{3} \int_0^6 \int_0^{3 - \frac{1}{2}x} (\frac{2}{3}x - 4 - \frac{8}{3}y) dydx \\ &= \frac{1}{3} \int_0^6 \left[(\frac{2}{3}x - 4)(3 - \frac{1}{2}x) - \frac{4}{3}(3 - \frac{1}{2}x)^2 \right] dx \\ &= \frac{1}{3} \int_0^6 \left[-\frac{2}{3}x^2 + 8x - 24 \right] dx \\ &= \frac{1}{3} \left[-\frac{2}{9}x^3 + 4x^2 - 24x \right]_0^6 = -\frac{48}{3} = -16 . \end{aligned}$$

Therefore the Stokes theorem is satisfied.

(b) Verify the Stokes theorem for Question 4(b).

(i) We found in Q4(b) that

$$\iint_{\mathcal{S}} \vec{F} \cdot \vec{n} dS = 4\pi \,.$$

(ii) Now, notice that the surface $\mathcal{S}$ has boundary the circle

$$\mathcal{C} : \begin{cases} x^2 + y^2 = 4, \\ z = 4 \end{cases} \quad \text{with a counterclockwise oritentation.}$$

Hence, $\mathcal{C}$ is described by the vector function

$$\vec{r}(t) := (2\cos t, 2\sin t, 4) \text{ with } t \in [0, 2\pi] \,.$$

Hence, we obtain that

$$\begin{aligned}
\oint_{\mathcal{C}} \vec{F} \cdot d\vec{r} &= \int_0^{2\pi} (4, 2\cos t, 4\sin^2 t) \cdot (-2\sin t, 2\cos t, 0)\, dt \\
&= \int_0^{2\pi} (-8\sin t + 4\cos^2 t)\, dt = \int_0^{2\pi} [-8\sin t + 2(\cos(2t) + 1)]\, dt \\
&= \Big[4\cos t + \sin(2t) + 2t\Big]_0^{2\pi} = 4\pi \,.
\end{aligned}$$

Hence, the Stokes theorem is verified.